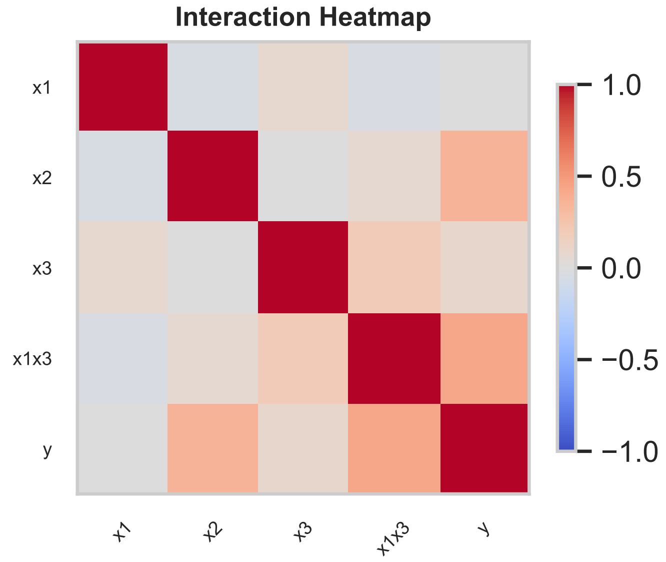
Data

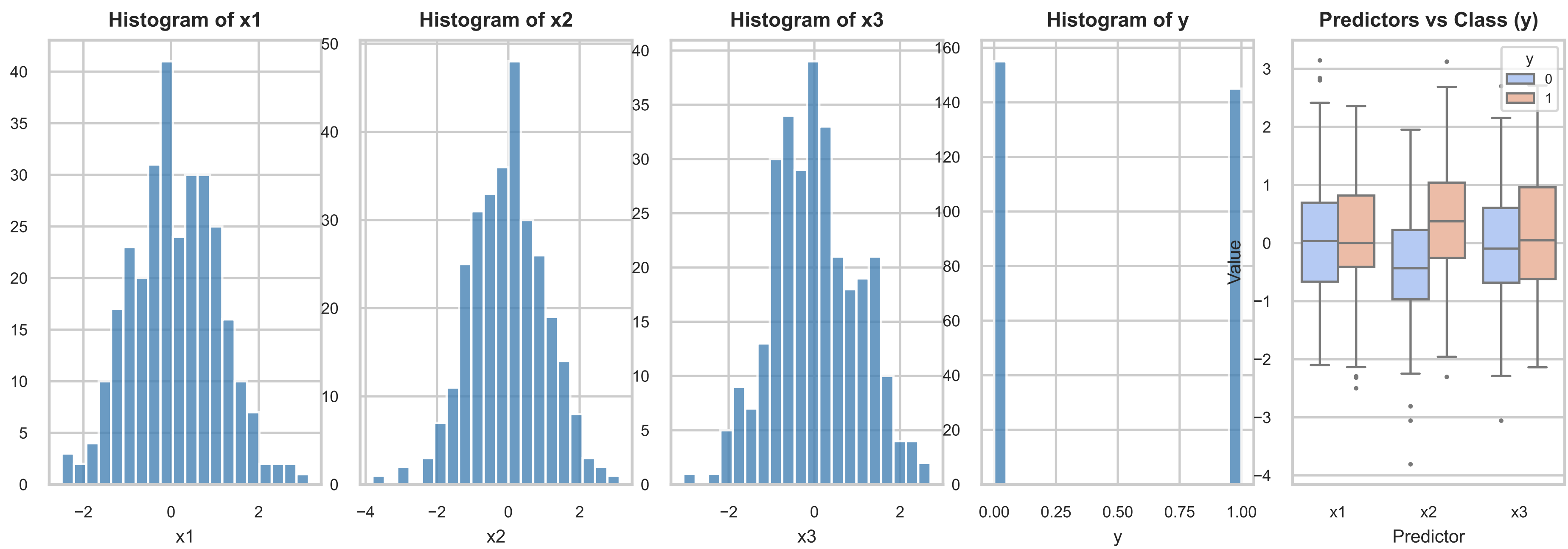
Sample Size & Structure

```
Training observations: 300
Test observations: 100
Predictors: x1, x2, x3, x1x3
Response: y (Binary Classification)
Class 1 Balance: 48.3%
```

Summary Statistics

	x1	x2	x3	y
mean	0.10	-0.02	0.04	0.48
std	0.99	1.05	1.00	0.50
min	-2.50	-3.81	-3.06	0.00
max	3.15	3.12	2.71	1.00





Model, Method and Results

Model & Assumptions

```
Logistic Model:
  Logit(p) = b0 + b1*x1 + b2*x2
             + b3*x3 + b4*(x1*x3)
  where p = P(y=1)

Assumptions:
  1. Linearity of the log-odds
  2. Independence of observations
  3. Lack of Multicollinearity
  4. Proper specification (Interaction)
```

Method

```
Method:
  Logistic Regression
  Maximum Likelihood Est. (MLE)
  QDA

 Library / Package:
  statsmodels (sm.Logit)
  sklearn
  pandas & numpy (data handling)
```

Logic & Strategy

```
Strategy:
  - x2: Retained as primary linear driver.
  - x1 & x3: Included to capture the
    XOR-style interaction term.

Marginality Principle:
  Main effects (x1, x3) were kept in
  the model to support the higher-order
  interaction term (x1*x3).
```

Model Formula

```
Estimated Equation (Log-Odds):
  z = -0.184 + 0.038*x1
      + 0.901*x2 + 0.028*x3
      + 1.538*(x1*x3)

Classification Rule:
  y_hat = 1 if [1 / (1 + exp(-z))] > 0.5
  else 0
```

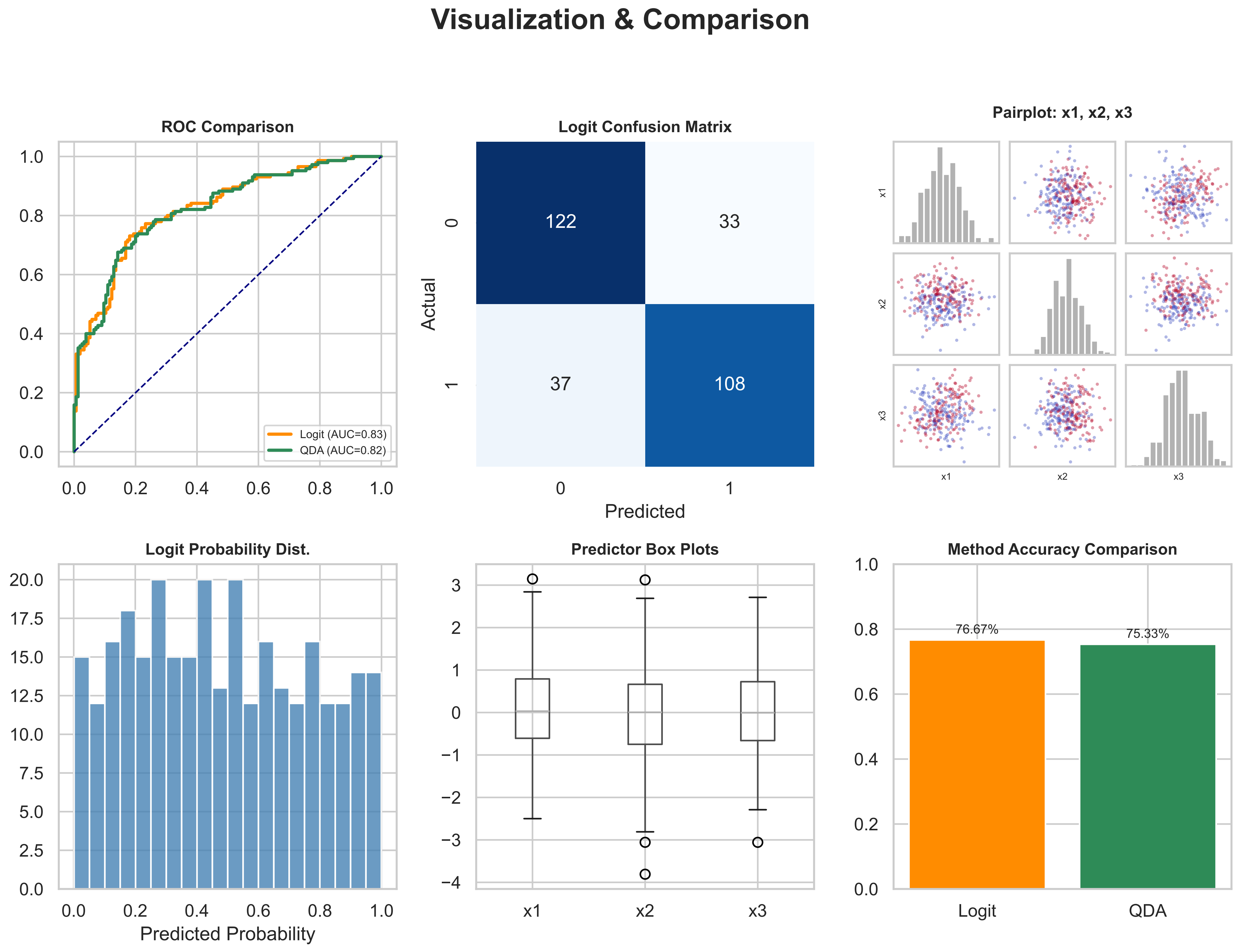
Coefficients (z-test)

Name	Coef	SE	p
const	-0.184	0.142	0.193
x1	0.038	0.161	0.816
x2	0.901	0.153	0.000***
x3	0.028	0.165	0.867
x1*x3	1.538	0.243	0.000***

Performance Summary

```
Performance Summary:
  Accuracy: 76.67%
  Log-Loss: 0.5074
  Pseudo R2: 0.2674

Optimization:
  Converged: True
  Iterations: 7
  LLR p-value: 4.16e-23
```


Visualization & Comparison
ROC Comparison
Logit (AUC=0.83)
QDA (AUC=0.82)
Logit Confusion Matrix
Actual
Predicted
122
33
37
108
Pairplot: x1, x2, x3
x1
x2
x3
Logit Probability Dist.
Predicted Probability
Predictor Box Plots
Method Accuracy Comparison
76.67%
75.33%
Logit
QDA